

References

- Edgar D Adrian. The impulses produced by sensory nerve endings: Part 1. *The Journal of Physiology*, 1926.
- Edgar D Adrian and Yngve Zotterman. The impulses produced by sensory nerve-endings: Part 2. the response of a single end-organ. *The Journal of Physiology*, 1926a.
- Edgar D Adrian and Yngve Zotterman. The impulses produced by sensory nerve endings: Part 3. impulses set up by touch and pressure. *The Journal of Physiology*, 1926b.
- Sanjeev Arora, Zhiyuan Li, and Kaifeng Lyu. Theoretical analysis of auto rate-tuning by batch normalization. *arXiv preprint arXiv:1812.03981*, 2018.
- Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016.
- Thomas Bachlechner, Bodhisattwa Prasad Majumder, Henry Mao, Gary Cottrell, and Julian McAuley. Rezero is all you need: Fast convergence at large depth. In *UAI*, 2021.
- Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli. wav2vec 2.0: A framework for self-supervised learning of speech representations. *NeurIPS*, 2020.
- Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, et al. Qwen technical report. *arXiv preprint arXiv:2309.16609*, 2023.
- David Balduzzi, Marcus Frean, Lennox Leary, JP Lewis, Kurt Wan-Duo Ma, and Brian McWilliams. The shattered gradients problem: If resnets are the answer, then what is the question? In *ICML*, 2017.
- Nils Bjorck, Carla P Gomes, Bart Selman, and Kilian Q Weinberger. Understanding batch normalization. *NeurIPS*, 2018.
- Andrew Brock, Soham De, and Samuel L Smith. Characterizing signal propagation to close the performance gap in unnormalized resnets. *arXiv preprint arXiv:2101.08692*, 2021a.
- Andrew Brock, Soham De, Samuel L Smith, and Karen Simonyan. High-performance large-scale image recognition without normalization. In *ICML*, 2021b.
- Zheng Cai, Maosong Cao, Haojiong Chen, Kai Chen, Keyu Chen, Xin Chen, Xun Chen, Zehui Chen, Zhi Chen, Pei Chu, et al. Internlm2 technical report. *arXiv preprint arXiv:2403.17297*, 2024.
- Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *ICCV*, 2021.
- Ekin D Cubuk, Barret Zoph, Jonathon Shlens, and Quoc V Le. Randaugment: Practical automated data augmentation with a reduced search space. In *CVPR Workshops*, 2020.
- Yan Dai, Kwangjun Ahn, and Suvrit Sra. The crucial role of normalization in sharpness-aware minimization. *NeurIPS*, 2024.
- Hadi Daneshmand, Jonas Kohler, Francis Bach, Thomas Hofmann, and Aurelien Lucchi. Batch normalization provably avoids ranks collapse for randomly initialised deep networks. *NeurIPS*, 2020.
- Soham De and Sam Smith. Batch normalization biases residual blocks towards the identity function in deep networks. *NeurIPS*, 2020.
- Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *CVPR*, 2009.
- Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Leo Feng, Frederick Tung, Mohamed Osama Ahmed, Yoshua Bengio, and Hossein Hajimirsadegh. Were rnns all we needed? *arXiv preprint arXiv:2410.01201*, 2024.

- Foundation Model Stack. Github: FMS FSDP. <https://github.com/foundation-model-stack/fms-fsdp>. Accessed: 2025-01-23.
- Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Alain Le Noac'h, Haonan Li, Kyle McDonell, Niklas Muennighoff, Chris Ociepa, Jason Phang, Laria Reynolds, Hailey Schoelkopf, Aviya Skowron, Lintang Sutawika, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. A framework for few-shot language model evaluation. <https://zenodo.org/records/10256836>.
- Leo Gao, Stella Biderman, Sid Black, Laurence Golding, Travis Hoppe, Charles Foster, Jason Phang, Horace He, Anish Thite, Noa Nabeshima, Shawn Presser, and Connor Leahy. The Pile: An 800gb dataset of diverse text for language modeling. *arXiv preprint arXiv:2101.00027*, 2020.
- Xinyang Geng and Hao Liu. Openllama: An open reproduction of llama, 2023. https://github.com/openlm-research/open_llama.
- Ensembl GRCh38. p13 (genome reference consortium human build 38), insdc assembly, 2013.
- Katarína Grešová, Vlastimil Martinek, David Čechák, Petr Šimeček, and Panagiotis Alexiou. Genomic benchmarks: a collection of datasets for genomic sequence classification. *BMC Genomic Data*, 2023.
- Albert Gu and Tri Dao. Mamba: Linear-time sequence modeling with selective state spaces. *arXiv preprint arXiv:2312.00752*, 2023.
- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, et al. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.
- Baran Hashemi, Roderic G Corominas, and Alessandro Giacchetto. Can transformers do enumerative geometry? *arXiv preprint arXiv:2408.14915*, 2024.
- HazyResearch. Github: Hyenadna. <https://github.com/HazyResearch/hyena-dna.git>. Accessed: 2025-01-23.
- Bobby He and Thomas Hofmann. Simplifying transformer blocks. *arXiv preprint arXiv:2311.01906*, 2023.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, 2016.
- Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *CVPR*, 2022.
- Stefan Heimersheim. You can remove gpt2’s layernorm by fine-tuning. *arXiv preprint arXiv:2409.13710*, 2024.
- Gao Huang, Yu Sun, Zhuang Liu, Daniel Sedra, and Kilian Q Weinberger. Deep networks with stochastic depth. In *ECCV*, 2016.
- Lei Huang, Xianglong Liu, Yang Liu, Bo Lang, and Dacheng Tao. Centered weight normalization in accelerating training of deep neural networks. In *ICCV*, 2017.
- Lei Huang, Jie Qin, Yi Zhou, Fan Zhu, Li Liu, and Ling Shao. Normalization techniques in training dnns: Methodology, analysis and application. *TPAMI*, 2023.
- Xiao Shi Huang, Felipe Perez, Jimmy Ba, and Maksims Volkovs. Improving transformer optimization through better initialization. In *ICML*, 2020.
- Hugging Face. Hugging Face: LLaMA 2. <https://huggingface.co/meta-llama/Llama-2-7b-hf>. Accessed: 2025-01-23.
- Sergey Ioffe and Christian Szegedy. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *ICML*, 2015.
- Nandan Kumar Jha and Brandon Reagen. Aero: Softmax-only llms for efficient private inference. *arXiv preprint arXiv:2410.13060*, 2024.
- Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.
- Ryo Karakida, Shotaro Akaho, and Shun-ichi Amari. The normalization method for alleviating pathological sharpness in wide neural networks. *NeurIPS*, 2019.

- Günter Klambauer, Thomas Unterthiner, Andreas Mayr, and Sepp Hochreiter. Self-normalizing neural networks. *NeurIPS*, 2017.
- Kuleshov Group. Github: Caduceus. <https://github.com/kuleshov-group/caduceus.git>. Accessed: 2025-01-23.
- Yann LeCun. A path towards autonomous machine intelligence version 0.9.2, 2022-06-27. *Open Review*, 2022.
- Pengxiang Li, Lu Yin, and Shiwei Liu. Mix-ln: Unleashing the power of deeper layers by combining pre-ln and post-ln. *arXiv preprint arXiv:2412.13795*, 2024.
- Aixin Liu, Bei Feng, Bin Wang, Bingxuan Wang, Bo Liu, Chenggang Zhao, Chengqi Deng, Chong Ruan, Damai Dai, Daya Guo, et al. Deepseek-v2: A strong, economical, and efficient mixture-of-experts language model. *arXiv preprint arXiv:2405.04434*, 2024.
- Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *CVPR*, 2022.
- Ilya Loshchilov, Cheng-Ping Hsieh, Simeng Sun, and Boris Ginsburg. ngpt: Normalized transformer with representation learning on the hypersphere. *arXiv preprint arXiv:2410.01131*, 2024.
- Ekdeep S Lubana, Robert Dick, and Hidenori Tanaka. Beyond batchnorm: Towards a unified understanding of normalization in deep learning. *NeurIPS*, 2021.
- Kaifeng Lyu, Zhiyuan Li, and Sanjeev Arora. Understanding the generalization benefit of normalization layers: Sharpness reduction. *NeurIPS*, 2022.
- Meta Research. Github: ConvNeXt. <https://github.com/facebookresearch/ConvNeXt>, a. Accessed: 2025-01-23.
- Meta Research. Github: DINO. <https://github.com/facebookresearch/dino>, b. Accessed: 2025-01-23.
- Meta Research. Github: DiT. <https://github.com/facebookresearch/DiT>, c. Accessed: 2025-01-23.
- Meta Research. Github: MAE. <https://github.com/facebookresearch/mae>, d. Accessed: 2025-01-23.
- Meta Research. Github: wav2vec 2.0. <https://github.com/facebookresearch/fairseq>, e. Accessed: 2025-01-23.
- Maximilian Mueller, Tiffany Vlaar, David Rolnick, and Matthias Hein. Normalization layers are all that sharpness-aware minimization needs. *NeurIPS*, 2024.
- Eric Nguyen, Michael Poli, Marjan Faizi, Armin Thomas, Michael Wornow, Callum Birch-Sykes, Stefano Massaroli, Aman Patel, Clayton Rabideau, Yoshua Bengio, et al. Hyenadna: Long-range genomic sequence modeling at single nucleotide resolution. *NeurIPS*, 2024.
- Toan Q Nguyen and Julian Salazar. Transformers without tears: Improving the normalization of self-attention. *arXiv preprint arXiv:1910.05895*, 2019.
- Yunhao Ni, Yuxin Guo, Junlong Jia, and Lei Huang. On the nonlinearity of layer normalization. *arXiv preprint arXiv:2406.01255*, 2024.
- Vassil Panayotov, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. Librispeech: an asr corpus based on public domain audio books. In *ICASSP*, 2015.
- William Peebles and Saining Xie. Scalable diffusion models with transformers. In *ICCV*, 2023.
- Siyuan Qiao, Huiyu Wang, Chenxi Liu, Wei Shen, and Alan Yuille. Micro-batch training with batch-channel normalization and weight standardization. *arXiv preprint arXiv:1903.10520*, 2019.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *JMLR*, 2020.
- Tim Salimans and Durk P Kingma. Weight normalization: A simple reparameterization to accelerate training of deep neural networks. *NeurIPS*, 2016.
- Shibani Santurkar, Dimitris Tsipras, Andrew Ilyas, and Aleksander Madry. How does batch normalization help optimization? *NeurIPS*, 2018.
- Yair Schiff, Chia-Hsiang Kao, Aaron Gokaslan, Tri Dao, Albert Gu, and Volodymyr Kuleshov. Caduceus: Bi-directional equivariant long-range dna sequence modeling. *arXiv preprint arXiv:2403.03234*, 2024.
- Jie Shao, Kai Hu, Changhu Wang, Xiangyang Xue, and Bhiksha Raj. Is normalization indispensable for training deep neural network? *NeurIPS*, 2020.